

SHIMIZU, Japan

WMO: 478980

Lat: 32.7231N

Lon: 133.0103E

Elev: 34

StdP: 100.92

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.7	3.0	-8.2	1.9	5.3	-6.9	2.1	5.7	9.9	14.2	8.5	13.2	3.3	0	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	31.0	26.4	30.2	26.2	29.6	25.9	27.3	29.6	26.9	29.2	26.5	28.8	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	28.9	26.2	21.8	28.6	25.8	21.2	28.3	86.9	29.7	85.0	29.2	83.1	28.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	7.9	6.9	-0.5	32.8	1.2	1.0	-1.4	33.5	-2.1	34.1	-2.8	34.6	-3.7	35.3		
			-2.8	28.3	1.2	0.8	-3.6	28.9	-4.3	29.4	-5.0	29.9	-5.8	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	8.8	9.8	12.6	16.7	20.5	23.1	26.8	27.9	25.7	21.5	16.4	11.2
	DBStd	7.00	3.10	3.47	3.15	2.59	1.66	1.71	1.53	1.19	1.79	2.38	3.20	3.42
	HDD10.0	144	61	43	12	1	0	0	0	0	0	0	1	27
	HDD18.3	1077	296	240	179	60	3	0	0	0	0	4	74	221
	CDD10.0	3229	23	37	92	201	325	392	520	556	471	357	193	64
	CDD18.3	1121	0	0	1	10	69	142	262	297	221	103	16	0
	CDH23.3	8527	0	0	0	2	67	507	2514	3337	1790	302	9	0
	CDH26.7	1903	0	0	0	0	2	22	538	1021	308	13	0	0
Wind	WSAvg	3.4	3.4	3.5	3.7	3.6	3.4	3.5	3.3	3.3	3.4	3.4	3.4	3.4
Precipitation	PrecAvg	2465	92	113	175	233	253	336	226	247	339	233	134	75
	PrecMax	3676	364	329	415	451	496	602	696	973	1074	753	417	196
	PrecMin	1501	8	32	34	61	80	150	36	17	54	40	30	3
	PrecStd	505	74	68	85	95	101	119	137	192	201	148	82	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	20.1	21.2	23.3	26.1	28.4	32.0	32.2	30.9	28.1	24.7	20.7
		MCWB	16.0	17.2	16.8	18.5	19.3	25.0	26.0	26.6	26.2	24.4	21.8	17.5
	2%	DB	16.7	18.6	19.7	22.2	24.8	27.3	30.5	31.2	29.9	26.7	23.0	18.9
		MCWB	13.7	16.1	16.3	18.2	19.6	24.4	26.2	26.6	25.5	22.9	19.2	15.6
	5%	DB	14.9	17.1	18.7	21.4	24.0	26.4	29.7	30.6	29.1	25.8	22.1	17.4
		MCWB	11.4	14.0	15.3	17.7	19.8	23.9	26.0	26.4	25.0	22.2	18.4	13.7
	10%	DB	13.7	15.3	17.7	20.6	23.3	25.7	29.0	30.0	28.3	25.0	21.1	16.3
		MCWB	9.9	11.8	14.2	16.9	19.5	23.3	25.7	26.2	24.7	21.1	17.5	12.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	18.2	18.9	21.2	23.2	25.8	27.8	28.2	27.4	25.6	22.2	18.8
		MCDB	18.2	19.3	19.9	22.2	24.1	27.1	30.0	30.6	29.6	27.2	24.0	19.8
	2%	WB	14.6	16.5	17.4	20.1	22.2	25.2	27.1	27.6	26.6	24.4	20.5	16.3
		MCDB	16.3	18.2	18.9	21.4	23.3	26.5	29.3	29.9	28.7	26.1	22.2	18.2
	5%	WB	12.3	14.7	16.1	18.9	21.5	24.7	26.6	27.1	26.1	23.2	19.3	14.5
		MCDB	14.3	16.8	18.1	20.6	22.9	26.0	28.7	29.4	28.2	25.2	21.4	16.8
	10%	WB	10.4	12.4	14.8	17.8	20.6	24.0	26.1	26.8	25.6	22.0	18.2	13.1
		MCDB	13.1	14.8	17.2	20.0	22.4	25.5	28.2	29.1	27.7	24.3	20.6	15.9
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.3	6.3	5.7	5.0	3.9	3.6	4.0	4.3	4.8	5.5	6.0
		MCDBR	7.1	6.7	6.5	6.1	5.8	4.3	4.5	4.7	4.8	5.1	5.5	6.1
		MCWBR	6.9	6.9	6.3	5.3	4.8	3.2	2.9	3.0	3.4	4.2	5.0	5.7
	5% WB	MCDBR	6.6	6.5	5.8	5.0	4.4	3.4	4.0	4.2	4.3	4.3	4.8	5.8
		MCWBR	7.3	7.3	6.6	5.3	4.4	2.8	2.8	2.8	3.3	4.1	5.1	6.2
Clear-Sky Solar Irradiance	taub	0.400	0.444	0.500	0.524	0.534	0.547	0.534	0.512	0.485	0.460	0.419	0.390	
	taud	2.196	2.095	1.949	1.946	1.992	2.048	2.123	2.195	2.228	2.200	2.237	2.238	
	Ebn at Noon	793	795	778	780	775	761	769	782	786	771	768	780	
	Edh at Noon	117	143	179	187	180	170	157	144	134	128	112	107	
All-Sky Solar Radiation	RadAvg	2.55	3.15	4.18	4.89	5.13	4.30	5.46	5.54	4.39	3.46	2.67	2.31	
	RadStd	0.22	0.27	0.31	0.50	0.54	0.62	0.62	0.44	0.43	0.37	0.26	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page